

Final Exam Review Worksheet

Reminder: The final exam is cumulative!

1. Are each of the following statements true or false?

- (a) Every vector in $\mathbb{R}^2$ is the image of a vector under the transformation $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as

$$f(\mathbf{u}) = \begin{bmatrix} 4 & 2 \\ -1 & 3 \end{bmatrix} \mathbf{u}$$

- (b) The distance between the vectors $\mathbf{u}$ and $\mathbf{v}$ is the length/norm of their difference.
- (c) If $\mathbf{v}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{w}$, then $\mathbf{v}$ is orthogonal to every vector in $\text{span}\{\mathbf{u}, \mathbf{w}\}$.
- (d) If $T : V \rightarrow W$ is a linear transformation with matrix representation $M_{DB}(T)$ for ordered bases B and D , and $\det(M_{DB}) = 5$, then T is an isomorphism.
- (e) A linearly independent set in an inner product space is also an orthonormal set.
- (f) The length/norm of the vector $\mathbf{v}$ is $\mathbf{v} \cdot \mathbf{v}$.
- (g) $T : P_1 \rightarrow P_2$ by $T(p(x)) = (p(x))^2$ is a linear transformation.
- (h) An orthonormal set can contain the zero vector.
- (i) If $T : \mathbb{R}^4 \rightarrow M_{2 \times 3}$ is a linear transformation represented by the matrix $M_{DB}(T)$, $M_{DB}(T)$ is a 4×6 matrix.
- (j) Using the inner product $\langle p, q \rangle = \int_0^1 p(x)q(x)dx$ for P_1 , the distance between $p(x) = x + 1$ and $q(x) = 3$ is $\sqrt{5}$.
- (k) $M_{3 \times 4}$ and P_{11} are isomorphic vector spaces.
- (l) If $M_B(L)$ and $M_D(L)$ are two matrix representations for the same linear operator $L : V \rightarrow V$, then $M_B(L)$ and $M_D(L)$ have the same determinant.
- (m) Let $L : V \rightarrow W$ be a linear transformation. If $\mathbf{v}_1$ and $\mathbf{v}_2$ are in $\ker L$, then so is $\text{span}\{\mathbf{v}_1, \mathbf{v}_2\}$.
- (n) If $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ is a linear transformation, then it is possible that $\dim \ker T = 1$ and $\dim \text{im} T = 2$.

2. Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the matrix transformation defined as

$$f\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 1 & 2 & 3 \\ -3 & -2 & -1 \\ -2 & 0 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

- (a) Find a basis for the image of f .

- (b) Is there a vector $\mathbf{x}$ such that $f(\mathbf{x}) = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$? Why or why not?

3. Let $L : P_1 \rightarrow P_1$ be a linear transformation defined by $L(x - 1) = x + 2$, $L(x + 1) = 2x + 1$.

- (a) What is $L(5x + 1)$?
- (b) What is $L(ax + b)$?

4. Let $L : P_2 \rightarrow P_2$ be a linear transformation defined by $L(ax^2 + bx + c) = (a + c)x^2 + (b + c)x$.

- (a) Is $x^2 - x - 1$ in the kernel of L ?
- (b) Is $x^2 + x - 1$ in the kernel of L ?
- (c) Is $2x^2 - x$ in the image of L ?
- (d) Is $x^2 - x + 2$ in the image of L ?
- (e) Find a basis for the kernel of L .
- (f) Find a basis for the image of L .

5. Let $L : M_{2 \times 2} \rightarrow \mathbb{R}$ be defined by

$$L\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = a - d + b - c.$$

Is L a linear transformation? Explain.

6. Let $L : M_{2 \times 2} \rightarrow M_{2 \times 2}$ be defined by $L(A) = A^T$.

Let

$$B = \left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right)$$

and

$$D = \left(\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)$$

be ordered bases for $M_{2 \times 2}$. Find:

- (a) $M_B(L)$
- (b) $M_{DB}(L)$
- (c) $M_{BD}(L)$
- (d) $M_D(L)$
- (e) Using $M_B(L)$ find the eigenvalues and associated eigenvectors of L .
- (f) Show that one of the eigenspaces is the set of all 2×2 symmetric matrices, and the other is the set of all 2×2 skew-symmetric matrices.

7. Let $L : P_2 \rightarrow P_2$ be the linear transformation defined by
 $L(ax^2 + bx + c) = (a + 2c)x^2 + (b - c)x + (a - c)$.

Let $B = (1, x, x^2)$ and $D = (x^2 - 1, x, x - 1)$ be ordered bases for P_2 .

- (a) Find $M_{DB}(L)$.
- (b) If $p(x) = 2x^2 - 3x + 1$, compute $L(p(x))$ using the matrix obtained in part (a).
- (c) Is L one-to-one?
- (d) Is L onto?

8. Let $L : P_1 \rightarrow P_1$ be a linear transformation represented by the matrix

$$M_B(L) = \begin{bmatrix} 2 & -3 \\ 1 & 2 \end{bmatrix}$$

with $B = (x - 2, x + 1)$.

- (a) Compute $L(x - 2)$ and $L(x + 1)$.
- (b) Compute $C_B(L(x - 2))$ and $C_B(L(x + 1))$.
- (c) Compute $L(x + 2)$.
- (d) Is L an isomorphism?

9. Let $L : P_1 \rightarrow P_1$ be the linear operator defined by $L(ax + b) = \frac{a + b}{2}x$. Let $B = (2 - x, 3 + x)$ be a basis for P_1 .

- (a) Find $C_B(L(2 - x))$ and $C_B(L(3 + x))$.
- (b) Find $M_B(L)$.
- (c) Find the eigenvalues and associated eigenvectors of $M_B(L)$.
- (d) Find the eigenvalues and associated eigenvectors of L .
- (e) Describe the eigenspace for each eigenvalue of L .
- (f) Is L diagonalizable?
- (g) Is L an isomorphism?

10. Given the orthonormal basis $S = \left\{ \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} \right\}$ for $\mathbb{R}^3$ (using the standard dot product), write the vector $\mathbf{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ as a linear combination of the vectors in S .

11. Let V be P_1 , and consider the inner product $\langle f(x), g(x) \rangle = \int_0^1 f(x)g(x)dx$. Determine all polynomials in P_1 which are orthogonal to $p(x) = x + 1$.

12. Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 \\ 2 \\ -1 \\ -1 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

- (a) Show that S is an orthogonal set (using the standard dot product).
- (b) Determine an orthonormal set so that the $\text{span} S = \text{span} T$.

13. Consider $W =$ the set of 2×2 symmetric matrices (which is a subspace of $M_{2 \times 2}$), and let $T : P_2 \rightarrow W$ be $T(p(x)) = \begin{bmatrix} p(2) & p(0) + p(2) \\ p(0) + p(2) & p(0) \end{bmatrix}$.

- (a) Show that T is a linear transformation.
- (b) Using $B = (x^2, x, 1)$ and $D = \left(\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right)$ as ordered bases for P_2 and W respectively, build $M_{DB}(T)$.
- (c) Use $M_{DB}(T)$ to find a basis for the kernel of T .
- (d) Use $M_{DB}(T)$ to find a basis for the image of T .
- (e) Is T an isomorphism?